What is Probability

Alexander Shen

LIRMM / CNRS & University of Montpellier

Outline

Paradox of Probability Theory

Galton Board

Natural Sciences and Mathematics

Rolling Dice

More Examples

Predicting Unpredictable

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- tossing a coin

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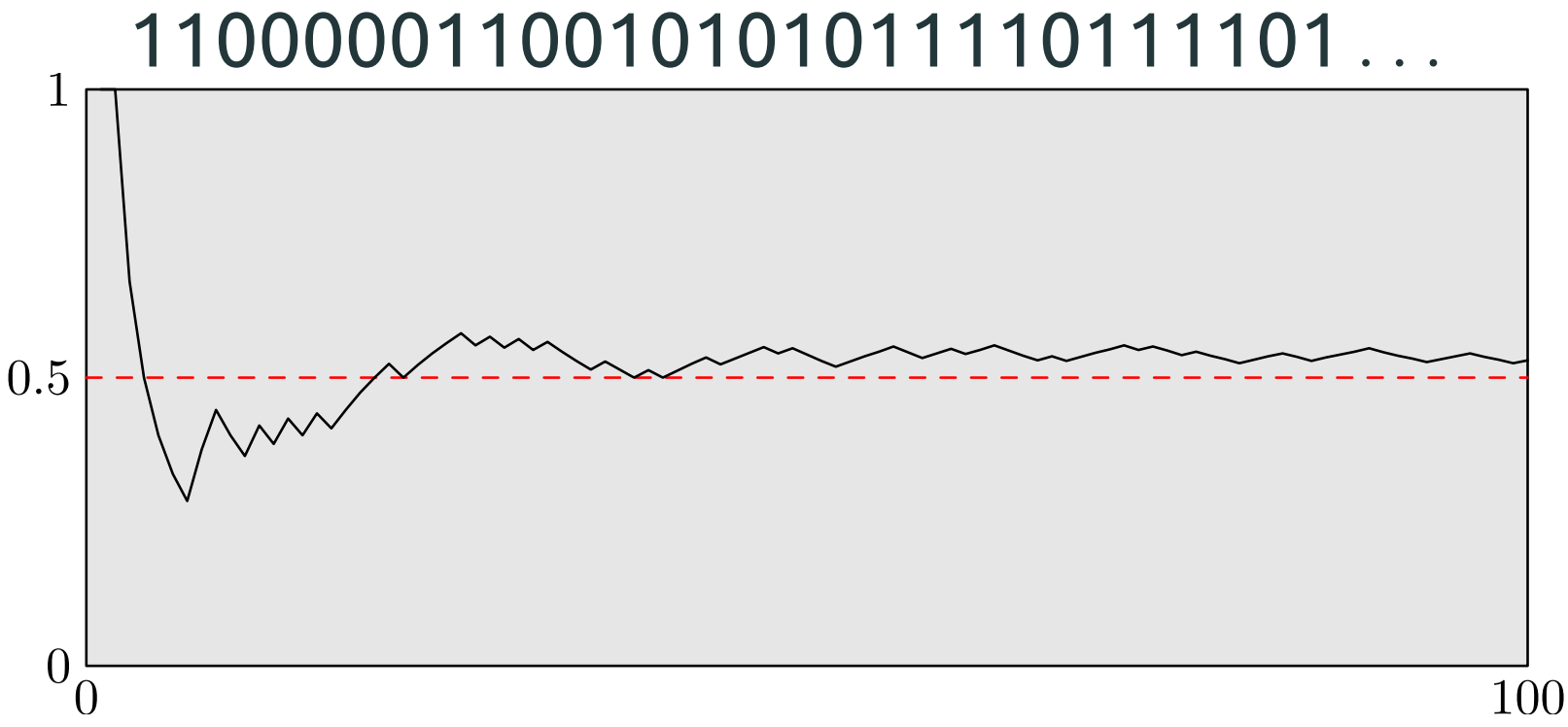
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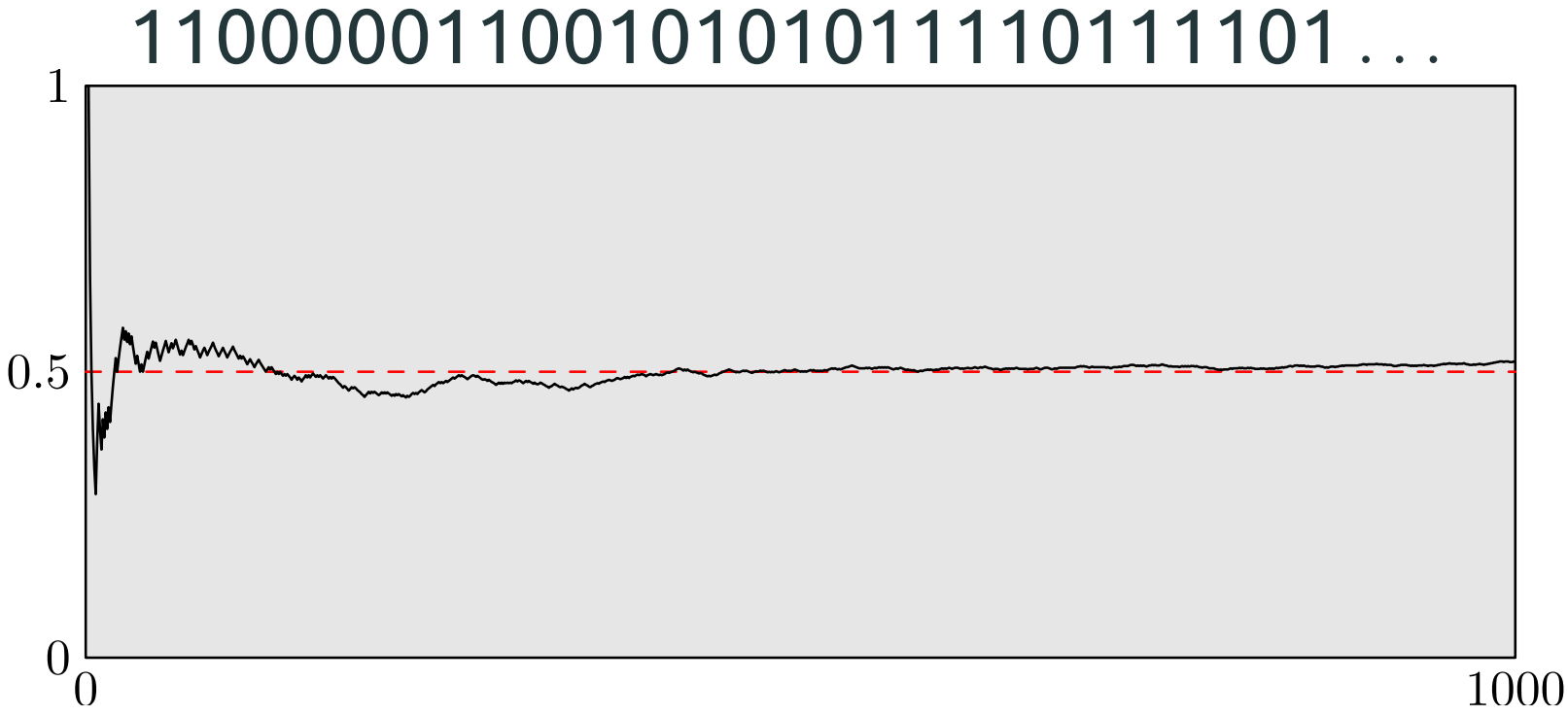
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- *frequency of 1s*: (#ones)/(length) $\approx 1/2$

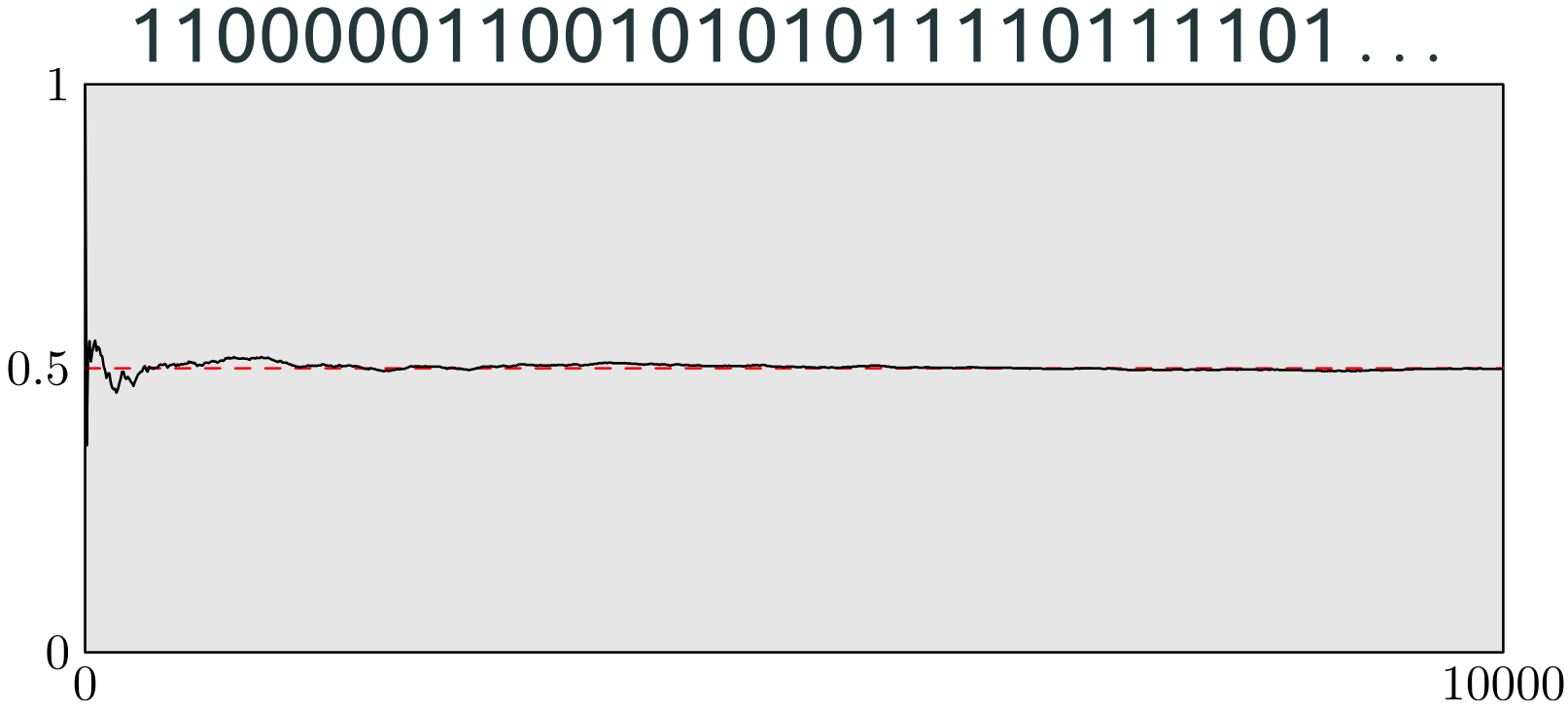
Hundred Random Bits



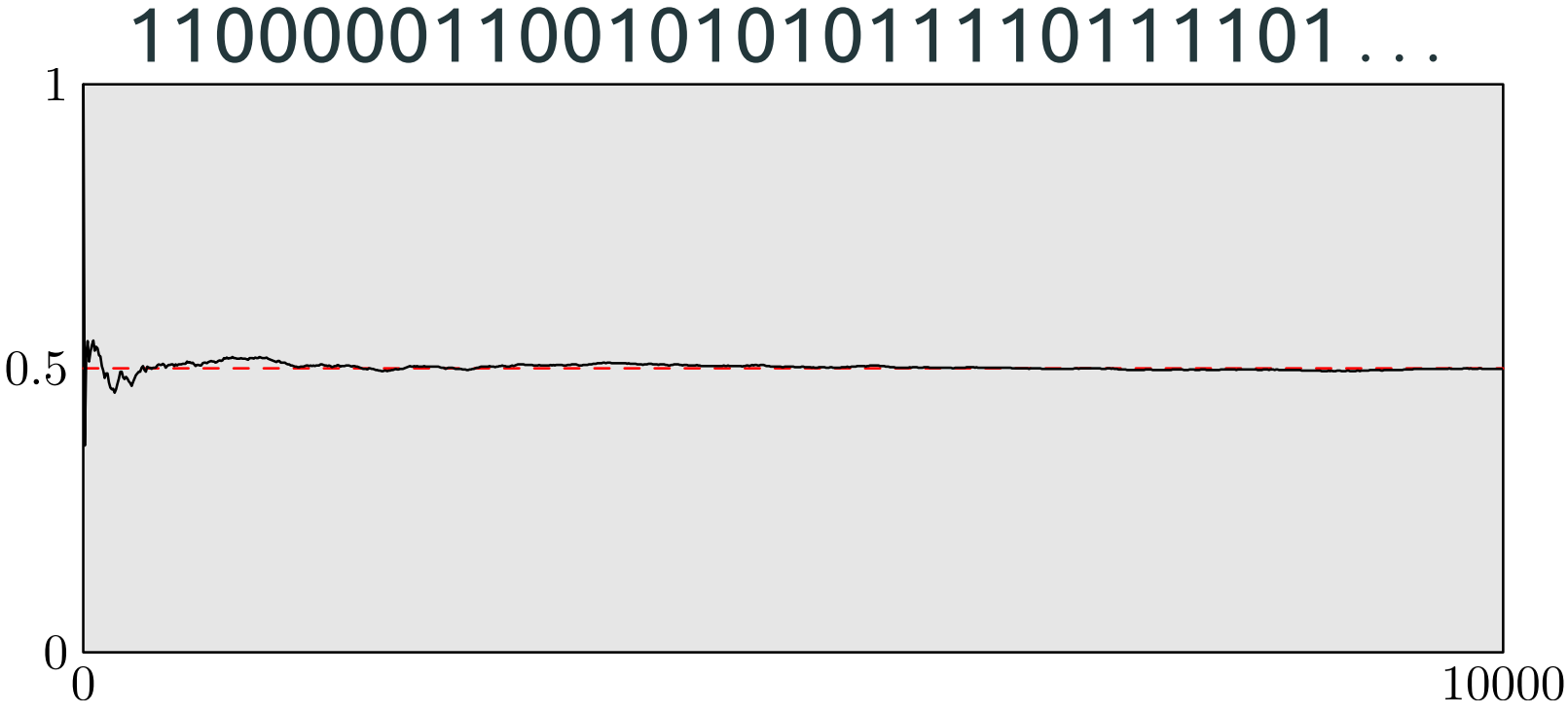
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Ten Thousands Random Bits



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(produced by George Marsaglia, 1995)

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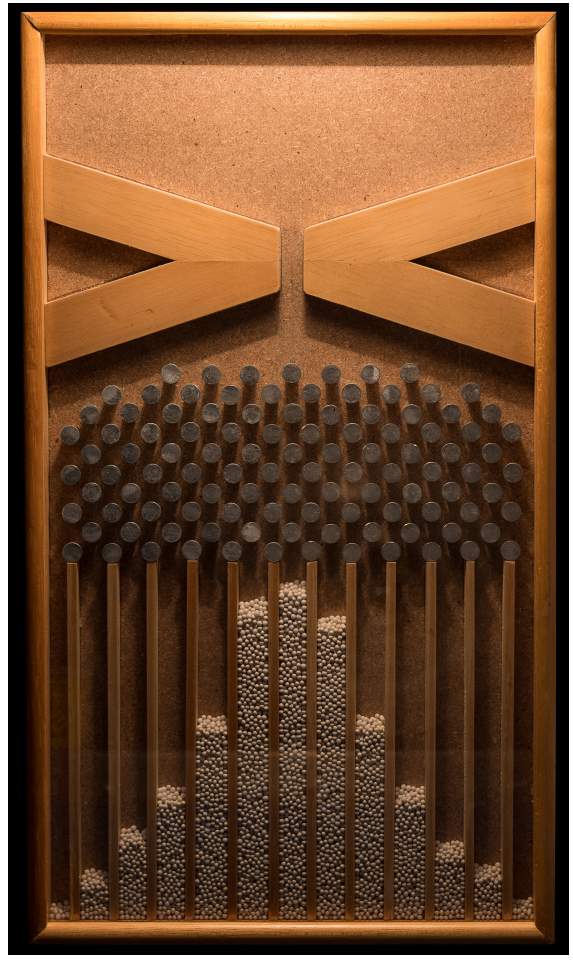
Galton Board

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More Examples

Bean Machine



https://en.wikipedia.org/wiki/Bean_machine

Sir Francis Galton (1822–1911)

Galton Board: animation

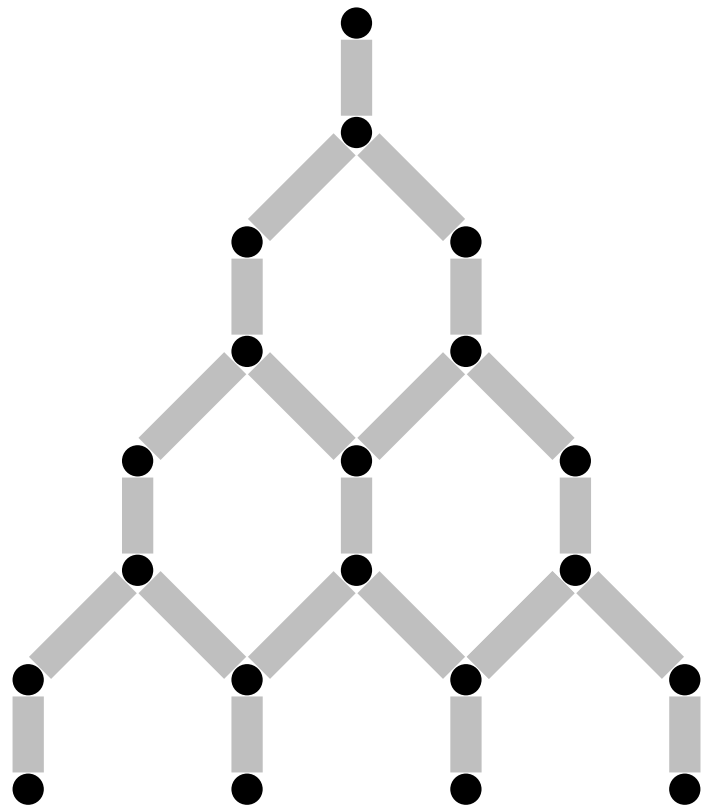
TO BE REPLACED BY VIDEO FROM

https://en.wikipedia.org/wiki/File:Galton_box.webm

Analysis

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Assume that at each level the beans are splitted evenly.



Galton and Pascal

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 & & & 1/2 & & & 1/2 & & & & \\
 & & 1/4 & & & 2/4 & & & 1/4 & & \\
 & 1/8 & & & 3/8 & & & 3/8 & & & 1/8 \\
 & & & & \ldots & & & & & &
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Pascal triangle/ 2^n: $\binom{n}{k}/2^n$

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- Dan: in fact, Alice is right for the real coins

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- among the beans that go left [right] at level 1, half go left [right] at level 2, etc.

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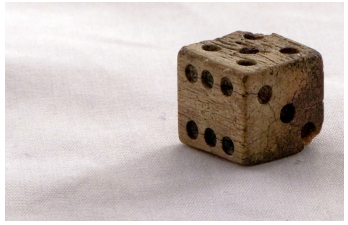
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Natural Sciences and Mathematics

Rolling Dice

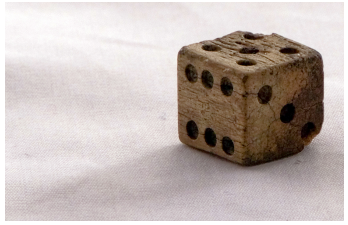
More Examples

Rolling a Dice



https://upload.wikimedia.org/wikipedia/commons/3/3b/Die_bone.jpg

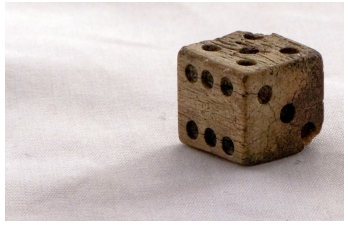
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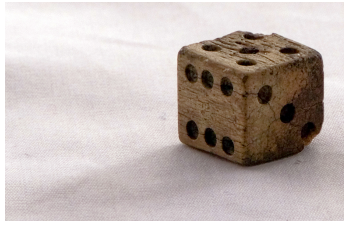
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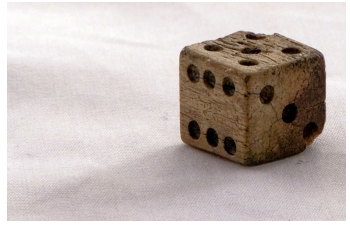
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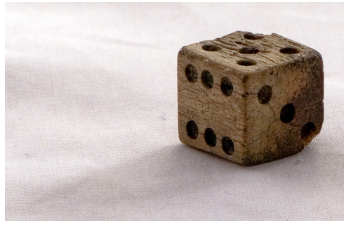
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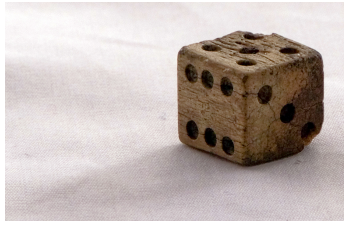
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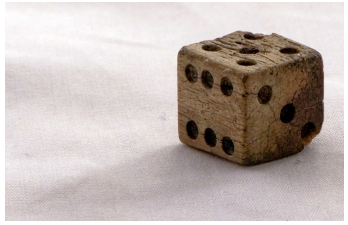
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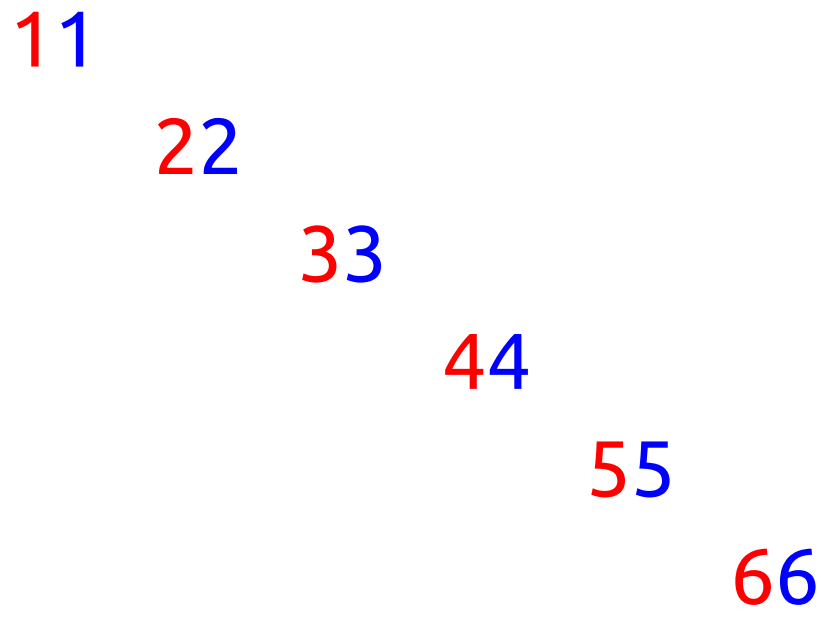
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- simultaneous and sequential setting
- equiprobable model is usually OK for both settings

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